

Haritha Hospitals — Comprehensive Guide

Audience: Processbricks team (Venkat + downstream devs) and any operator touching the Haritha Hospitals ERPNext stack. **Scope:** End-to-end reference for the Haritha Hospitals deployment — architecture, customizations, migration, ops, and lessons. **Last updated:** 2026-08-29 (post custom-app build + master data migration + P1 outage recovery).

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1. Overview

Field	Value
Project	Haritha Hospitals ERPNext deployment
Company	Haritha Hospitals (healthcare / hospital group)
Industry	Hospital shift management + HRMS
Owner	Venkat Narasimha (Processbricks)
Started	2026-04-01 (concept); project on this repo: 2026-08-19
Stack	Frappe 16 / ERPNext 16 / HRMS 16.5.0 (pinned) / custom app haritha_hospital
Sites	pberpprod.duckdns.org (prod), pberpdev.duckdns.org (dev), dev-erp.duckdns.org (Venkat VPS for prototyping)
Last major milestone	2026-08-29 — custom app build + master data migration + P1 outage recovery

Goal: Stand up Haritha Hospitals as a real hospital shift-management system backed by ERPNext + HRMS, capturing all customizations as a portable Frappe custom app so any env can be reproduced with `bench install-app haritha_hospital`.

Out of scope (deferred): Wards, beds, OTs, pharmacy, lab, billing, full Chart of Accounts, cost centers beyond the 2 already created. Haritha adds these modules incrementally once shift + HRMS is validated.

2. Tech Stack

Core Apps (4)

App	Version	Notes
Frappe	v16.30.0 (pinned per compose)	Foundation framework
ERPNext	v16.30.0	Accounting, inventory, selling, buying
HRMS	v16.5.0 (pinned, see LEARNINGS #44)	HR, payroll, shift, attendance, leave. v16.5.1+ breaks on repost_allowed_types — never upgrade without testing on dev first.
Custom app	haritha_hospital (0.0.1)	Owns 274 customizations as fixtures

Infrastructure

- **MariaDB** 10.x via Docker named volumes
- **Docker Compose** (compose-based deployment; `erp-{env}-*` containers on main VPS, `erpdev-*` on Venkat VPS)
- **Redis** (cache + queue broker) + **Socket.IO** for realtime desk
- **nginx-proxy** reverse proxy with TLS termination (DuckDNS + Let's Encrypt / self-signed fallback)
- **Custom Python 3.11** (Frappe v16 baseline)

Custom App: haritha_hospital

- **Repo:** `git@github.com:venkat-narasimha/haritha_hospital.git`
- **Branch:** `main`
- **Layout:** Standard Frappe app — git-root with inner Python module dir (`apps/haritha_hospital/haritha_hospital`)
- **Captures:** 274 customizations as JSON fixtures (78 Custom Field + 189 Property Setter + 3 Print Format + 2 Notification + 2 Letter Head)
- **Hooks:** `fixtures = [...]` block lists 5 DocTypes (see `HARITHA_HOSPITALS_GUIDE §5`)

Source Repos

- **App:** `https://github.com/venkat-narasimha/haritha_hospital`
- **Workspace / docs / scripts:** `https://github.com/venkat-narasimha/haritha_hospitals` (this repo)
- **Memory / lessons (separate, not in this repo):** `/root/.openclaw/workspace/MEMORY.md` + `/root/.openclaw/workspace/.learnings/LEARNINGS.md`

3. Architecture

3.1 Environment Layout

Env	URL	Compose Project	Containers	DB Name	Purpose
pberpprod	https://pberpprod.duckdns.org/	erp-prod	9 (backend, frontend, scheduler, queue-short, queue-long, websocket, redis-cache, redis-queue, db)	_b80f05e76a0d	Production. Customizations + 8,118 Shift Assignments.
pberpdev	https://pberpdev.duckdns.org/	erp-dev	9 (same layout)	_0c0679ad719b	Development. Mirror of prod customizations + master data.
dev-erp	https://dev-erp.duckdns.org/	erpdev	9 (different prefix erpdev-*)	_f042e7eabf48	Mat VPS prototyping. v15 HRMS, separate stack.

QA env (pberppqa.duckdns.org) exists but **skipped for Haritha** per Venkat 2026-08-28 — direct dev → prod promotion with custom-app fixtures as the safety net.

3.2 Customization Capture

```
pberpprod DB
- 274 customizations (CF + PS + PF + N + LH)
- 8,118 Shift Assignments + 6,300 Attendance
- 210 Employees + 25 Shift Types + ...
```

```
bench export-fixtures
(requires fixtures = [...] in hooks.py)
```

```
haritha_hospital app
apps/haritha_hospital/haritha_hospital/
  fixtures/
    custom_field.json (78 rows)
    property_setter.json (189 rows)
    print_format.json (3 rows)
    notification.json (2 rows)
    letter_head.json (2 rows)
  hooks.py ← fixtures = [...] declaration
```

```
git push → GitHub
bench install-app haritha_hospital
bench --site <env> migrate
```



Any new env
Same 274 customizations, env-portable, idempotent

3.3 Master Data Layer

Master data (Company, Employee, Department, Shift Type, etc.) is **NOT in fixtures**. Rationale: - Master data is environment-specific (test fixtures would clutter prod) - Master data drifts frequently (new hires, terminations) — fixtures would force re-export churn - Migration is a one-shot operation per env, not a continuous sync

Master data lives in the DB; CSV sources in `projects/haritha-hospitals/masters/` (19 files, ~1.77 MB) act as the canonical reference. Migration script: `scripts/migrate_master_data.py` (idempotent).

3.4 Persistence Model

Volume	Container	Content	Recovery on restart
erp-{env}_sites	backend, frontend, scheduler, workers	/home/frappe/frappe-bench/sites/ (site_config.json, apps.txt, assets, private files)	□ preserved
erp-{env}_db-data	db only	/var/lib/mysql/ (MariaDB data dir)	□ preserved
erp-{env}_redis-cache-data	redis-cache	cache snapshots	□ preserved
erp-{env}_redis-queue-data	redis-queue	background job queue	□ preserved
Anonymous (writable layer)	any	runtime writes (logs, tmp)	△ ephemeral

Rule of thumb: anything in `/home/frappe/frappe-bench/` is on a named volume. docker restart is 100% safe; docker-compose down -v is data loss.

4. Phase History

Phase 0 — Schema Planning (2026-04-01 to 2026-08-19)

Goal: Document schema for all master data entities needed for shift management.

Status: □ Complete. 19 entities, 168 fields. HRMS v15 verification done (9 docs read, 7 corrections applied).

Deliverables: - `all_schemas.csv` — 19 entities, 168 fields - `TRACKER.md` — this repo's tracker - `README.md` — project overview - `knowledge/shift_management_hrms.md` — reusable reference

Decisions: - 2026-04-01: Scope = shift management only (defer wards, beds, OTs, pharmacy, lab, billing) - 2026-04-15: HRMS v16.5.0 pin (Lesson #44) - 2026-08-19: Shift code = 10-char

[P][HHMM][S][HHMM] (HRMS-native flags) - 2026-08-19: Holidays = standard Indian national + 4-5 Telangana

Phase 1 — Schema Approval (2026-08-20)

Goal: Manager sign-off on schema + 19 data CSVs.

Status: ☒ Done. Both deliverables on GitHub (private repo). Manager downloaded, imported to Google Sheets, shared with team.

Phase 1.5 — CSV Master Re-Verification (2026-08-25)

Goal: Validate 19 CSV masters before ingestion.

Status: ☒ Complete. All 7 checks PASS. 0 FAILs, 0 WARNs. 24,758 rows across 19 entities.

Findings (MEMORY candidates): - Actual shift code format = [GMAN]\d{4}[RS]\d{4} (10-char) — MEMORY decision was wrong (said [P][HHMM][S][HHMM]) - employee.csv has NO name column — FKjoin key = attendance_device_id (EMP-NNNN) - Verify script format assumption: ## Data marker + header + data rows; use csv.DictReader on header slice

Phase 2 — Site Setup (2026-08-25, after 2026-08-21 rollback)

Goal: Spin up Haritha Hospitals on pberpprod.duckdns.org with clean state.

Status: ☒ Done. Pre-flight backup + wipe + reinit executed.

Apps installed: frappe, erpNext, hrms 16.5.0 (pinned), payments. Custom app **deferred** initially.

Phase 3 — Data Import (2026-08-26)

Goal: Load 19 CSV masters into prod.

Final tally: 24,511 records across 9 entities (Company, Holiday, Department, Designation, Employment Type, Shift Type, Employee, Shift Assignment, Attendance, Employee Checkin).

Key gotchas: - Employee PK = HR-EMP-NNNNN (autoname), not CSV EMP-NNNN. Map via employee_number. - Attendance status extended with 'Holiday' + 'Weekly Off' via Property Setter - Synthetic data: defaults for gender (Not Specified), date_of_birth (1990-01-01), first_name (split from employee_name) - 5 X-HH Department variants force-deleted via direct SQL (Frappe doc.delete() enforces disable-not-delete) - HR-Attendance series counter fixed mid-flight (was 183, fixed to 6300)

Phase 3.5 — Reconcile + SS/SSA/SR Synthesis (2026-08-26)

Status: ☒ Done. All 11 entities match CSV targets after dedup + re-ingest.

- Department: 47 → 37 (36 CSV + 'All Departments' root)
- Designation: 76 → 48
- Leave Type: 9 → 7
- Employment Type: 8 → 6 (merged CSV-added with 3 defaults)
- Holiday: 28 → 14
- Shift Location: 1 → 0 (deleted bogus '(no rows)' literal)
- SS/SSA/SR synthesized to fill Phase 3 deferral gap

Phase 3.6 — Bulk Submit (2026-08-27)

Status: ☐ Done. 6,314 docs at docstatus=0 now docstatus=1.

3 runs needed due to 3-layer Frappe framework barriers (LEARNINGS #106): 1. Naming series backfill (raw SQL ingest missed naming_series) 2. Property Setter for Attendance status options (added 'Holiday' + 'Weekly Off') 3. Monkey-patch controller-level status check (HRMS Attendance.validate() has hardcoded 5-value list)

Phase 3.7 — Property Setter Recreate Script (2026-08-27)

Status: ☐ Done. scripts/recreate_property_setters.py is idempotent.

Pattern: scripted recreate (Option 2) instead of fixtures, because HRMS' hooks.py doesn't list 'Property Setter' as a fixture (LEARNINGS #151). Mirrors bulk_submit.py's importlib.util.spec_from_file_location() pattern.

Phase 4 — Roster Crash Fix (2026-08-28)

Issue: HRMS Roster page returned 500 on pberpprod.

Root cause: CapitalCase color keys ('Blue') vs lowercase Tailwind palette ('blue') in apps/hrms/roster/src/components/MonthViewTable.vue. TypeScript color type was lowercase but data had CapitalCase.

Fix: Property Setters for Shift Type color options + default + lowercase color values across all 25 Shift Types (78 → 25 visible colors actually stored; 50 missing shifts regenerated with defaults).

Lesson: "Worked before" + "Works on other site" = data bug, not framework.

Phase 4.10/4.11 — Roster Real Root Cause + Verification (2026-08-28)

Deep-dive verification via headless Chromium with auth cookie injection (CDP Network.setCookie). Confirmed Phase 4.10 was the correct fix; no new code change needed, just documentation + idempotent verification script.

Lessons: #147-#150 (minified Vue column positions, numeric vs string keys, CDP HttpOnly cookies, verification scripts).

Custom App Build (2026-08-28)

Goal: Capture all customizations as Frappe fixtures for env portability.

Built haritha_hospital app, configured hooks.py, exported 274 customizations, installed on both pberpdev (fresh) + pberpprod (idempotent).

Schema gotchas caught: - module column vs legacy app (LEARNINGS #152) - fixtures = [...] declaration required (LEARNINGS #151) - Letter Head has no module column (LEARNINGS #155) - Stock Print Formats have module=NULL (LEARNINGS #156)

Master Data Migration (2026-08-29)

16 DocTypes migrated from prod to dev. 8,118 Shift Assignments on dev (7,829 from prod + 289 pre-existing).

Migration script: scripts/migrate_master_data.py (idempotent, 16 DocTypes, 10 gotchas documented — LEARNINGS #157).

P1 Outage — Both envs 500'd (2026-08-29 03:06 IST)

Root cause: Gunicorn --preload froze sys.path at container startup; today's haritha_hospital install added .pth but gunicorn didn't see it.

Fix: docker restart erp-{dev,prod}-backend-1 after volume verify (Lesson #72). ~30-60s downtime each, zero data loss.

Lesson: Always restart backend container after bench install-app (LEARNINGS #153).

Phase 5 — DEFERRED (2026-08-28, per Venkat)

DR test, error monitoring, security audit, SSL audit, UAT, perf baseline — all skipped.

Phase 6 — TBD

ISO/CMM L5 docs originally planned after Phase 5. Awaiting Venkat.

5. Customizations Catalog (274)

5.1 Custom Fields (78)

Major DocTypes with new fields:

DocType	New Fields
Employee	PAN, IFSC, bank account, Health Insurance Number, approvers (Leave Approver, Expense Approver, Shift Request Approver)
Company	Payroll cost center, default leave policy
Attendance	HRMS-specific status extensions, late entry/early exit flags
Shift Type	Color (Tailwind palette), HRMS workflow flags
Shift Assignment	Department link, custom status
Leave Application	Approval workflow tweaks
Holiday List	Telangana regional holidays
...	(full list in apps/haritha_hospital/haritha_hospital/fixtures/cu

5.2 Property Setters (189)

Largest category. Dominant DocTypes:

DocType	PS Count	Purpose
HRMS (Attendance, Shift Type, Shift Assignment, Employee, Leave Application)	~120	Status options, default values, mandatory toggles, field order
ERPNext (Company, Item, Customer, Supplier)	~50	Custom field visibility, validation rules
Frappe core (User, File, Communication)	~19	UI behavior tweaks

5.3 Print Formats (3)

- Haritha Hospitals Payslip
- Haritha Hospitals Shift Card
- Haritha Hospitals Leave Application

5.4 Notifications (2)

- **DISABLED:** Shift assignment change notification
- **DISABLED:** Leave approval pending notification

5.5 Letter Heads (2)

- Haritha Hospitals (default for all docs)
- Haritha Hospitals - Confidential (for HR/Payroll)

5.6 Total = 78 + 189 + 3 + 2 + 2 = 274 □

6. Migration Guide (How to Replicate)

6.1 Setup New Env with Customizations

1. Spin up the env (compose-based)

```
cd /path/to/frappe_docker
docker compose -p erp-newenv up -d
```

2. Create site

```
docker exec erp-newenv-backend-1 bash -c "cd /home/frappe/frappe-bench && \
  bench new-site newsitename --admin-password admin123 --mariadb-root-password <root>"
```

3. Install base apps

```
docker exec erp-newenv-backend-1 bash -c "cd /home/frappe/frappe-bench && \
  bench --site newsitename install-app erpnext"
docker exec erp-newenv-backend-1 bash -c "cd /home/frappe/frappe-bench && \
  bench --site newsitename install-app hrms --branch version-16"
```

4. Get haritha_hospital custom app

```
docker exec erp-newenv-backend-1 bash -c "cd /home/frappe/frappe-bench && \
  bench get-app https://github.com/venkat-narasimha/haritha_hospital"
```

5. Install + restart (CRITICAL – see LEARNINGS #153)

```
docker exec erp-newenv-backend-1 bash -c "cd /home/frappe/frappe-bench && \
  bench --site newsitename install-app haritha_hospital"
docker restart erp-newenv-backend-1
```

6. Verify count

```
docker exec erp-newenv-backend-1 bash -c "cd /home/frappe/frappe-bench && \
  bench --site newsitename console < /tmp/verify_haritha.py"
# Expect: 78 CF + 189 PS + 3 PF + 2 N + 2 LH = 274
```


6.2 Migrate Master Data

Use scripts/migrate_master_data.py (idempotent, 16 DocTypes):

```
docker cp scripts/migrate_master_data.py erp-newenv-backend-1:/tmp/  
docker exec erp-newenv-backend-1 bash -c "cd /home/frappe/frappe-bench && \  
  bench --site newsitename console < /tmp/run_migrate.py"  
# where run_migrate.py imports + calls migrate_all()
```

6.3 Migration Gotchas (LEARNINGS #157)

1. frappe.get_doc() requires doctype key in payload
2. Company default accounts depend on Account existing first → 2-step insert + update
3. Account root nodes (Accounts Receivable, etc.) are system-managed → skip
4. Shift Type autoname='prompt' → set name explicitly
5. Department/Item Group root (All Departments) → skip
6. Employee missing Gender/Default Shift → migrate lookup data first
7. Employee ID remapping — prod IDs differ from dev IDs, look up by employee_name
8. Child tables stripped by REST API fields=["*"] — need explicit fetch
9. HRMS validate_approver for Shift Request → Department Approver must exist
10. autoname='field.#####' requires Series pre-bump or Frappe renames

6.4 Pre-Flight Checklist Before Any Migration

- [] Backup taken (4 files: DB, config, public, private)
 - [] Backup SHA256 verified
 - [] Source env custom app count = 274
 - [] Target env custom app count = 274 (after install-app)
 - [] Backend container restarted after install-app (LEARNINGS #153)
 - [] All named volumes intact (Lesson #72 verify)
 - [] Migration script dry-run passes
 - [] Reversal plan documented (drop + restore from backup)
-

7. Operational Runbook

7.1 Daily

- ☐ Verify cron backup slot (4x/day per env) — check offsite rsync success
- ☐ Read backup log tail (NOT just exit code — Lesson #114)
- ☐ Spot-check container health (docker ps | grep erp-)
- ☐ Monitor token burn (HEARTBEAT.md §7)

7.2 Weekly

- ☐ Verify all envs healthy (/ returns 200 on prod + dev)
- ☐ Check git repo sync (git status clean + origin/main reachable)
- ☐ Review open lessons in .learnings/ for new patterns

7.3 Monthly

- ☐ **Re-verify DB passwords against container env** (LEARNINGS #154): for env in dev prod qa; do docker exec erp-\${env}-db-1 printenv MYSQL_ROOT_PASSWORD; done
- ☐ Audit MEMORY.md for stale hardcoded values

- ☐ Audit LEARNINGS.md for stale entries
- ☐ Test restore from backup (3-2-1 rule, Lesson #4)
- ☐ Review cron logs for partial failures

7.4 Per Change (Customization, Migration, Update)

- ☐ **Pre-flight checklist** (§6.4 above)
- ☐ Apply on dev first, verify, then prod
- ☐ After bench install-app: **docker restart erp-{env}-backend-1** (LEARNINGS #153)
- ☐ After any DB write: backup before + verify after (Lesson #72)
- ☐ Export new customizations to fixtures + commit + push (Rule #9)
- ☐ Update TRACKER.md + this guide

7.5 Incident Response: HTTP 500 on All Requests

If both/all envs suddenly return HTTP 500:

1. Check if recent `bench install-app` ran → likely gunicorn --preload issue (LEARNINGS #153)
 2. Verify container uptime vs last install-app time
 3. Volume verify (Lesson #72): `docker inspect <container> | grep Mounts`
 4. If volume verify passes → `docker restart erp-<env>-backend-1`
 5. If volume verify fails → STOP, do not restart, escalate
-

8. Open Issues / Phase 6 Backlog

8.1 Deferred from Today (2026-08-29)

- **Prod admin password recovery** — pberpprod login returned 401 post-restart; need to retrieve actual ADMIN_PASSWORD from container .env (MEMORY's PUNJD5Hmp5B0uWBkRgeo literal also drifted)
- **HRMS version-check** in haritha_hospital/__init__.py — should enforce hrms == 16.5.0 to prevent accidental upgrades
- **MEMORY.md / LEARNINGS.md audit** — Lesson #114 silent cron failure root cause still has 3 dropped backup lines from a previous regression

8.2 Phase 5 Items Still Skipped (per Venkat 2026-08-28)

- Disaster recovery test (restore from backup)
- Error monitoring (Sentry / Rollbar integration)
- Security audit (HTTPS headers, RBAC review, password policy)
- SSL audit (cert expiry monitoring)
- UAT (user acceptance testing with Haritha Hospital staff)
- Performance baseline (load test, response time SLAs)

8.3 Phase 6 Candidates (TBD with Venkat)

- ISO 9001 + 27001 documentation
- SOPs + process maps
- Audit trail configuration
- Manager walkthrough deck
- Customer pilot (Haritha Hospital staff training)

8.4 Out-of-Scope Reminder

Wards, beds, OTs, pharmacy, lab, billing, full Chart of Accounts — deferred per Phase 0 decision. Add incrementally after shift + HRMS validated end-to-end.

9. References

Repos

- **App:** https://github.com/venkat-narasimha/haritha_hospital
- **Workspace / docs / scripts:** https://github.com/venkat-narasimha/haritha_hospitals (this repo)
- **OpenClaw workspace memory:** [git@github.com:venkat-narasimha/erpclaw.git](https://github.com/venkat-narasimha/erpclaw)

Frappe / ERPNext

- **ERPNext docs:** <https://docs.frappe.io/>
- **HRMS docs:** <https://docs.frappe.io/hr/>
- **Frappe framework:** <https://frappeframework.com/docs/>
- **Custom app tutorial:** <https://frappeframework.com/docs/v14/user/en/tutorial>

Internal

- **MEMORY.md:** `/root/.openclaw/workspace/MEMORY.md` (workspace root, separate repo)
- **LEARNINGS.md:** `/root/.openclaw/workspace/.learnings/LEARNINGS.md` (workspace root, separate repo)
- **Daily logs:** `/root/.openclaw/workspace/memory/YYYY-MM-DD.md`
- **Tracker:** `TRACKER.md` (in this repo)
- **Decisions log:** `DECISIONS.md` (in this repo)
- **Master data CSVs:** `masters/` (19 files, 1.77 MB)
- **Migration script:** `scripts/migrate_master_data.py`
- **Property Setter recreate script:** `scripts/recreate_property_setters.py`
- **Tracker update script:** `scripts/update_tracker.py`
- **CSV verification script:** `scripts/verify_csvs.py`

External

- **DuckDNS:** <https://www.duckdns.org/> (dynamic DNS for *.duckdns.org subdomains)
 - **Docker Compose:** <https://docs.docker.com/compose/>
 - **MariaDB:** <https://mariadb.com/kb/en/documentation/>
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Appendix A — Master Data State (as of 2026-08-29)

DocType	pberpprod count	pberpdev count	Source CSV
Company	1	1	masters/company.csv
Employee	210	210	masters/employee.csv
Department	37	37	masters/department.csv
Designation	48	48	masters/designation.csv

DocType	pberpprod count	pberpdev count	Source CSV
Account	89	89	(system defaults + 5 roots skipped)
Cost Center	2	2	(system + Haritha Hospitals)
Item Group	6	6	(system + Haritha)
UOM	239	239	(system + Haritha units)
Shift Type	25	25	masters/shift_type.csv
Holiday List	1	1	masters/holiday_list.csv
Holiday	14	14	masters/holiday.csv
Employment Type	6	6	masters/employment_type.csv
Gender	1 (Not Specified)	1	(system)
Shift Schedule	5	5	(synthesized)
Shift Assignment	5,318	8,118	masters/shift_assignment.csv + migration script
Shift Request	8	7	(synthesized + 1 migration conflict)
Leave Type	7	7	masters/leave_type.csv
Attendance	6,300	6,300	masters/attendance.csv
Employee Checkin	12,562	12,562	masters/employee_checkin.csv

Appendix B — Incident Timeline (P1 Outage, 2026-08-29)

Time (IST)	Event
~03:00	haritha_hospital install-app completed on both envs
03:06	Venkat reports: “both pberpdev and pberpprod are down - internal server error”
03:07	haritha-diag-500 subagent spawned (premium Minimax)
03:09	Root cause confirmed: gunicorn --preload sys.path freeze
03:11	Venkat asks “will restart clear the data?”
03:12	haritha-verify-volumes subagent spawned
03:18	Volume verify PASSES — all data on named volumes, zero risk
03:20	Venkat confirms YES → restart sequence spawned
03:21	docker restart erp-dev-backend-1 (30s)
03:22	dev curl verify: HTTP 200  , login 200 
03:23	docker restart erp-prod-backend-1 (30s)
03:24	prod curl verify: HTTP 200  , login 401  (DB password drifted)
03:30	Incident closed; MEMORY.md DB password audit kicked off
04:00	All action items documented in memory/2026-08-29.md

Total downtime: ~18 min (both envs combined).

Total data loss: zero.

Root cause documented in: LEARNINGS.md #153 (gunicorn preload), #154 (restart vs down).

Action items completed today: MEMORY.md DB password correction, LEARNINGS #151-#157 added, this guide created.

Action items still open: prod admin password recovery (401 incident), HRMS version-check in haritha_hospital/__init__.py, Phase 5 items (deferred per Venkat).

Last updated: 2026-08-29 09:21 IST (post Phase 0+ — Foundation & Migration).

Generated as part of comprehensive doc update subagent task (2026-08-29). See TRACKER.md Phase 0+ section for chronological build-out.